

```
/******
```

PS 8 : Control sintético
 Universidad de San Andrés
 Economía Aplicada
 Integrantes:

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```
*****/
```

```
global main "/Users/federicolopez/Library/CloudStorage/OneDrive-Personal/Documents/UDESA/08/
APLICADA/TUTORIALES/E178-APLICADA-PS-G1/T08/PS8"
```

```
global output "$main/output"
```

```
global input "$main/input"
```

```
cd "$main"
```

```
import delimited using "$main/input/df.csv", clear
replace state = "São Paulo" if state == "SÃo Paulo"
encode state, gen(state_n)
xtset state_n year
```

```
egen promedio_bra = mean(homiciderate) if state_n != 26, by(year)
```

```
*****
```

```
* fig 1
```

```
twoway ///
```

```
(line homiciderate year if state_n == 26, lwidth(thick) lcolor(black)) ///
```

```
(line promedio_bra year if state_n == 25, lpattern(dash)) ///
```

```
, ytitle("Homicide Rates") xtitle("Year") ///
```

```
ylabel(0(10)60) yscale(r(0 60)) ///
```

```
legend(label(1 "São Paulo") label(2 "Brazil (average)") ///
```

```
position(8) ring(0) cols(1) textwidth(23) symxsize(4)) ///
```

```
xline(1998, lpattern(dash) lcolor(gray))
```

```
graph export "$output/fig1.png", replace
```

```
synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln yearsschoolingimp giniimp
proportionextremepoverty, trunit(26) trperiod(1998) nested fig
mat list e(W_weights)
```

* En e(X_balance) se guarda la matriz con las medias de los predictores para el sintético y el verdadero

```
mat list e(X_balance)
```

```
* me guardo la serie sintetica
```

```
mat list e(Y_synthetic)
```

```
mat Y_synth = e(Y_synthetic)
```

```
gen Y_synth_valores = .
```

```
forvalues y = 1990/2009 {
```

```
    local sdfsd = Y_synth[`y' - 1989, 1]
```

```
    replace Y_synth_valores = `sdfsd' if year == `y'
```

```

}

*****

* fig 2
twoway ///
(line homiciderate year if state_n == 26, lwidth(thick) lcolor(black)) ///
(line Y_synth_valores year if state_n == 26, lpattern(dash)) ///
, ytitle("Homicide Rates") xtitle("Year") ///
ylabel(10(10)60) yscale(r(10 60)) ///
legend(label(1 "São Paulo") label(2 "Synthetic São Paulo") ///
position(8) ring(0) cols(1) textwidth(35) symxsize(4)) ///
xline(1998, lpattern(dash) lcolor(gray))
graph export "$output/fig2.png", replace

*****

* fig 3
gen gaps = .
replace gaps = (homiciderate - Y_synth_valores) if state_n == 26

twoway ///
(line gaps year if state_n == 26, lwidth(thick) lcolor(black)) ///
, ytitle("Gaps in Homicide Rates") xtitle("Year") ///
ylabel(-30(10)30) yscale(r(-30 30)) ///
xline(1998, lpattern(dash) lcolor(gray)) ///
yline(0, lpattern(dash) lcolor(black))
graph export "$output/fig3.png", replace

*****

* fig 4
import delimited using "$main/input/df.csv", clear
replace state = "São Paulo" if state == "São Paulo"
encode state, gen(state_n)
drop if year > 1998
xtset state_n year

synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln yearsschoolingimp giniimp
proportionextremepoverty, trunit(26) trperiod(1995) nested fig

mat Y_synth = e(Y_synthetic)
gen Y_synth_valores = .
forvalues y = 1990/1998 {
    local sdfsd = Y_synth[`y' - 1989, 1]
    replace Y_synth_valores = `sdfsd' if year == `y'
}

twoway ///
(line homiciderate year if state_n == 26, lwidth(thick) lcolor(black)) ///
(line Y_synth_valores year if state_n == 26, lpattern(dash)) ///

```

```
, ytitle("Homicide Rates") xtitle("Year") ///
ylabel(0(10)60) yscale(r(0 60)) ///
legend(label(1 "São Paulo") label(2 "Synthetic São Paulo") ///
position(8) ring(0) cols(1) textwidth(35) symxsize(4)) ///
xline(1995, lpattern(dash) lcolor(gray))
graph export "$output/fig4.png", replace
```

```
*****
```

```
* fig 5 : Leave one out
```

```
import delimited using "$main/input/df.csv", clear
replace state = "São Paulo" if state == "São Paulo"
encode state, gen(state_n)
xtset state_n year
tempname resmat
local i 26
```

```
qui synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln
yearsschoolingimp giniimp proportionextremepoverty, trunit(`i') trperiod(1998) xperiod(1990(1)2009)
keep(loo-resout`i', replace)
```

```
forvalues j=1/27 {
  if `j'==26 {
    continue
  }
  import delimited using "$main/input/df.csv", clear
  replace state = "São Paulo" if state == "São Paulo"
  encode state, gen(state_n)
  xtset state_n year
  * Remove a state based on encoded variable
  drop if state_n==`j'
  * Run synthetic control with updated data
  qui synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln
  yearsschoolingimp giniimp proportionextremepoverty, trunit(26) trperiod(1998) xperiod(1990(1)2009)
  keep(loo-resout`j', replace)
}
```

```
* volvemos a mergear las bases individuales
```

```
forvalues i = 1/27 {
  use "loo-resout`i'.dta", clear
  ren _Y_synthetic _Y_synthetic_`i'
  ren _Y_treated _Y_treated_`i'
  gen _Y_gap_`i'=_Y_treated_`i'-_Y_synthetic_`i'
  save "loo-resout`i'.dta", replace
}
```

```
use "loo-resout1.dta", clear
forvalues i = 2/27 {
```

```
merge 1:1 _Co_Number _time using "loo-resout`i'.dta", nogen
}
```

```
twoway (line _Y_synthetic_7 _time, lcolor(gray) lwidth(thin) ///
    lpattern(solid) legend(label(1 "Synthetic São Paulo (leave one out))) ) ///
(line _Y_synthetic_23 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_synthetic_24 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_treated_26 _time, ///
    lcolor(black) lpattern(solid) ///
    legend(label(4 "São Paulo") ) ) ///
(line _Y_synthetic_26 _time, ///
    lcolor(black) lpattern(dash) ///
    legend(label(5 "Synthetic São Paulo") lpattern(dash) ) ) , ///
ytlabel("Homicide Rates") xtitle("Year") ///
ylabel(10(10)60) yscale(r(10 60)) ///
legend(order(4 5 1) position(8) ring(0) cols(1) textwidth(60) symxsize(5)) ///
xline(1998, lpattern(dash) lcolor(gray))
```

```
graph export "$output/fig5.png", replace
```

```
*****
```

```
* fig 6 : placebos
```

```
clear all
import delimited using "$main/input/df.csv", clear
replace state = "São Paulo" if state == "SÃ£o Paulo"
encode state, gen(state_n)
xtset state_n year
```

```
tempname resmat
local i 26
qui synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln
yearsschoolingimp giniimp proportionextremepoverty, trunit(`i') trperiod(1998) xperiod(1990(1)2009)
keep(resout`i', replace)
* Resout es una matriz que nos guarda para cada "year": outcome sintetico, outcome treated, y
weights.
```

```
matrix`resmat' = nullmat(`resmat') \ e(RMSPE)
local names ``names' ``i'"""
mat colnames`resmat' = "RMSPE"
mat rownames`resmat' = `names'
matlist`resmat', row("Treated Unit")
drop if state_n == 26
```

```
forvalues i = 1/27 {
    if `i'==26 {
        continue
    }
    qui synth homiciderates stategdpcapita stategdpgrowthpercent populationprojectionln
```

```
yearsschoolingimp giniimp proportionextremepoverty, trunit(`i`) trperiod(1998) xperiod(1990(1)2009)
```

```
keep(resout`i`, replace)
```

```
matrix`resmat` = nullmat(`resmat`) \ e(RMSPE)
```

```
* Nos guardamos el MSPE
```

```
local names ``names' ``i'``
```

```
}
```

```
mat colnames`resmat` = "RMSPE"
```

```
mat rownames`resmat` = `names'
```

```
matlist`resmat`, row("Treated Unit")
```

```
* fijemonos que contiene cada base resout`i`
```

```
use "resout3.dta", clear
```

```
* renombro variables de cada base individual para despues hacer un merge.
```

```
forvalues i = 1/27 {
```

```
use "resout`i`.dta", clear
```

```
ren _Y_synthetic _Y_synthetic_`i`
```

```
ren _Y_treated _Y_treated_`i`
```

```
gen _Y_gap_`i`=_Y_treated_`i`-_Y_synthetic_`i`
```

```
save "resout`i`.dta", replace
```

```
}
```

```
use "resout1.dta", clear
```

```
forvalues i = 2/27 {
```

```
merge 1:1 _Co_Number _time using "resout`i`.dta", nogen
```

```
}
```

```
twoway (line _Y_gap_1 _time, lcolor(gray) lpattern(solid) ///
```

```
legend(label(1 "Control states"))) ///
```

```
(line _Y_gap_2 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_3 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_4 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_5 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_6 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_7 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_8 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_9 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_10 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_11 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_12 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_13 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_14 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_15 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_16 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_17 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_18 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_19 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_20 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```
(line _Y_gap_21 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
```

```

(line _Y_gap_22 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_23 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_24 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_25 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_27 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_26 _time, lcolor(black) lwidth(thick) lpattern(solid) legend(label(27 "São Paulo") ) ), ///
ytile("Gap in Homicide Rates") xtile("Year") ///
ylabel(-30(10)30) yscale(r(-30 30)) ///
legend(order(27 1) position(8) ring(0) cols(1) textwidth(30) symxsize(5)) ///
xline(1998, lpattern(dash) lcolor(gray)) ///
yline(0, lpattern(solid) lcolor(black))
graph export "$output/fig6.png", replace

```

* fig 7 : placebos (algunos)

```

clear all
import delimited using "$main/input/df.csv", clear
replace state = "São Paulo" if state == "São Paulo"
encode state, gen(state_n)
xtset state_n year

```

tempname resmat

local i 26

```

qui synth homiciderates stategdp capita stategdp growth percent population projection ln
yearsschooling imp gini imp proportion extreme poverty, trunit(`i') trperiod(1998) xperiod(1990(1)2009)
keep(resout`i', replace)

```

* Resout es una matriz que nos guarda para cada "year": outcome sintetico, outcome treated, y weights.

```
matrix `resmat' = nullmat(`resmat') \ e(RMSPE)
```

```
local names ``names' ``i'``''
```

```
mat colnames `resmat' = "RMSPE"
```

```
mat rownames `resmat' = `names'
```

```
matlist `resmat', row("Treated Unit")
```

```
drop if state_n == 26
```

```
forvalues i = 1/27 {
```

```
if `i'==26 {
```

```
continue
```

```
}
```

```

qui synth homiciderates stategdp capita stategdp growth percent population projection ln
yearsschooling imp gini imp proportion extreme poverty, trunit(`i') trperiod(1998) xperiod(1990(1)2009)
keep(resout`i', replace)

```

```
matrix `resmat' = nullmat(`resmat') \ e(RMSPE)
```

```
* Nos guardamos el MSPE
```

```
local names ``names' ``i'``''
```

```
}
```

```
mat colnames `resmat' = "RMSPE"
```

```

mat rownames `resmat' = `names'
matlist `resmat' , row("Treated Unit")

```

```
/*
```

```
Treated Unit | RMSPE
```

```
-----+-----
```

```
26 | 3.261704
```

```
1 | 7.052483
```

```
2 | 17.6648
```

```
3 | 14.71914
```

```
4 | 13.66277
```

```
5 | 2.948762 -
```

```
6 | 2.316661 -
```

```
7 | 17.27511
```

```
8 | 16.86609
```

```
9 | 4.127187 -
```

```
10 | 3.192369 -
```

```
11 | 12.30525
```

```
12 | 9.435383
```

```
13 | 10.26978
```

```
14 | 2.431522 -
```

```
15 | 13.18879
```

```
16 | 4.073998 -
```

```
17 | 23.21934
```

```
18 | 15.30902
```

```
19 | 3.835551 -
```

```
20 | 4.293679 -
```

```
21 | 30.58504
```

```
22 | 12.30056
```

```
23 | 19.78524
```

```
24 | 13.47249
```

```
25 | 10.09636
```

```
27 | 7.649329
```

```
*/
```

```
* RMSPE menores que 2x el de SP (<6.4):
```

```
* 5,6,9,10,14,16,19,20
```

```
* fijemonos que contiene cada base resout`i'
```

```
use "resout3.dta", clear
```

```
* renombro variables de cada base individual para despues hacer un merge.
```

```
forvalues i = 1/27 {
```

```
use "resout`i'.dta", clear
```

```
ren _Y_synthetic _Y_synthetic_`i'
```

```
ren _Y_treated _Y_treated_`i'
```

```
gen _Y_gap_`i'=_Y_treated_`i'-_Y_synthetic_`i'
```

```
save "resout`i'.dta", replace
```

```
}
```

```
use "resout1.dta", clear
forvalues i = 2/27 {
merge 1:1 _Co_Number _time using "resout`i'.dta", nogen
}
```

```
twoway (line _Y_gap_5 _time, lcolor(gray) lpattern(solid) ///
       legend(label(1 "Control states")) ) ///
(line _Y_gap_6 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_9 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_10 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_14 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_16 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_19 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_20 _time, lcolor(gray) lwidth(thin) lpattern(solid)) ///
(line _Y_gap_26 _time, lcolor(black) lwidth(thick) lpattern(solid) legend(label(9 "São Paulo") ) ), ///
ytile("Gap in Homicide Rates") xtile("Year") ///
ylabel(-30(10)30) yscale(r(-30 30)) ///
legend(order(9 1) position(8) ring(0) cols(1) textwidth(30) symxsize(5)) ///
xline(1998, lpattern(dash) lcolor(gray)) ///
yline(0, lpattern(solid) lcolor(black))
```

```
graph export "$output/fig7.png", replace
```